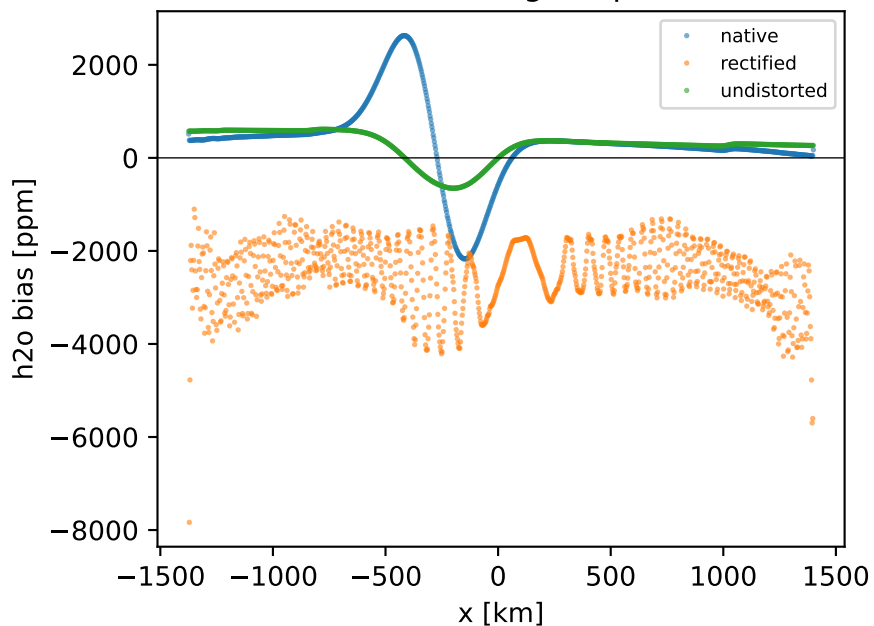
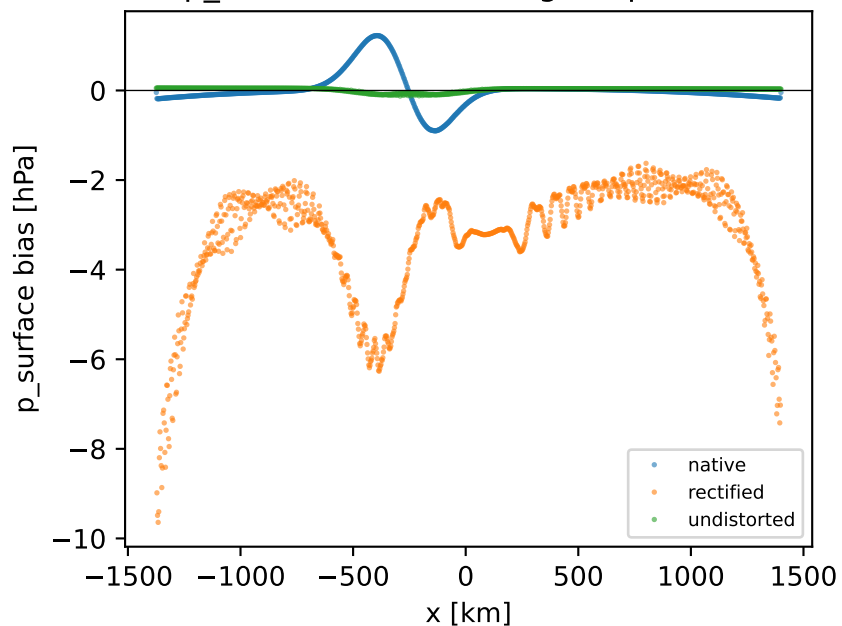


Joint FPA0 -- realistic, no noise (native/rectified/undistorted)

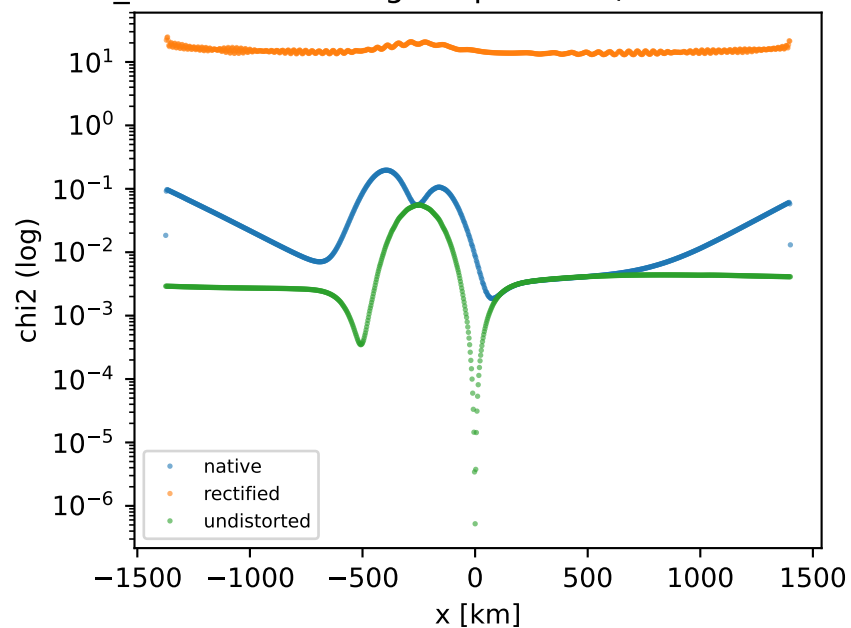
h2o bias vs along-slit position



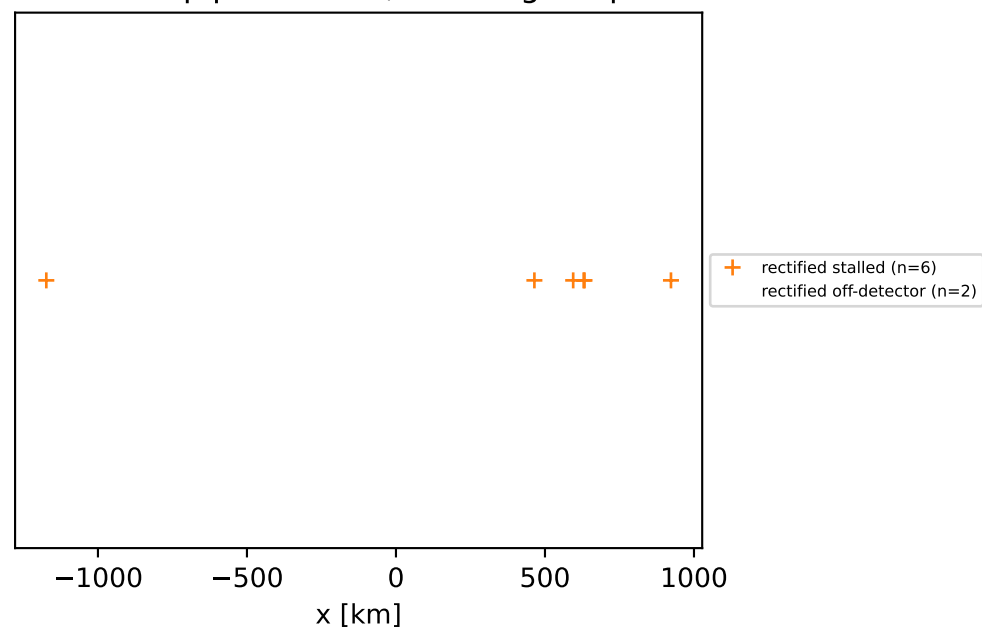
p_surface bias vs along-slit position



chi2_reduced vs along-slit position (outliers excluded)

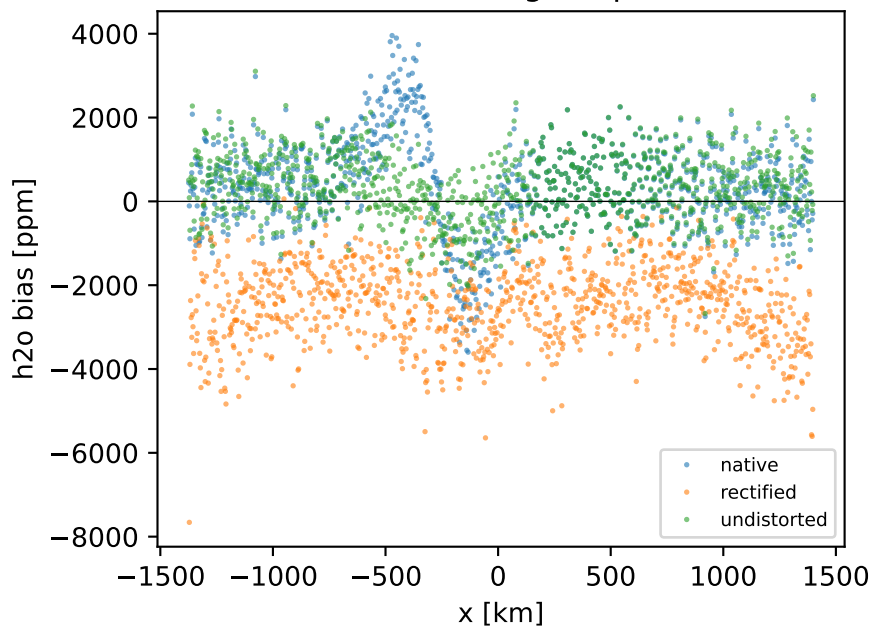


Where each pipeline fails, vs along-slit position

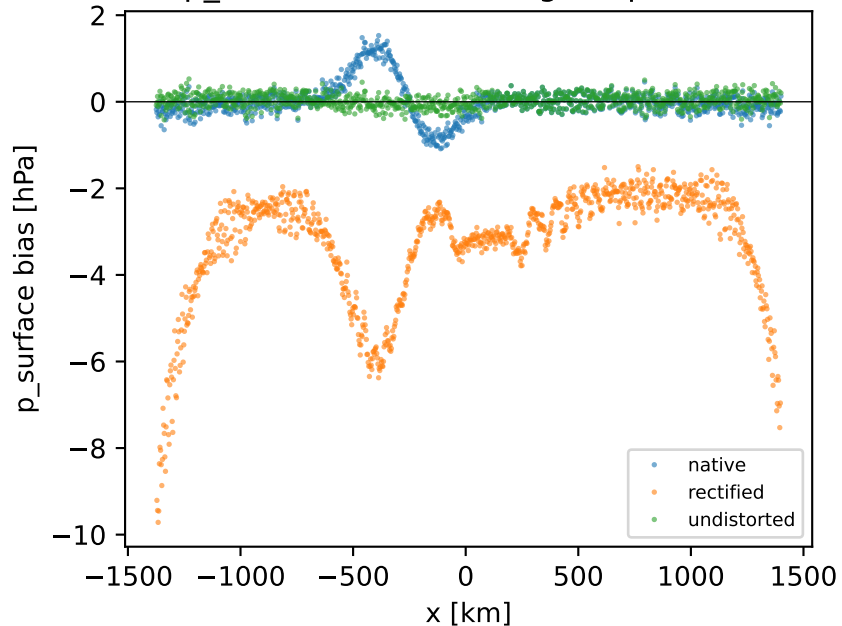


Joint FPA0 -- realistic, noise (native/rectified/undistorted)

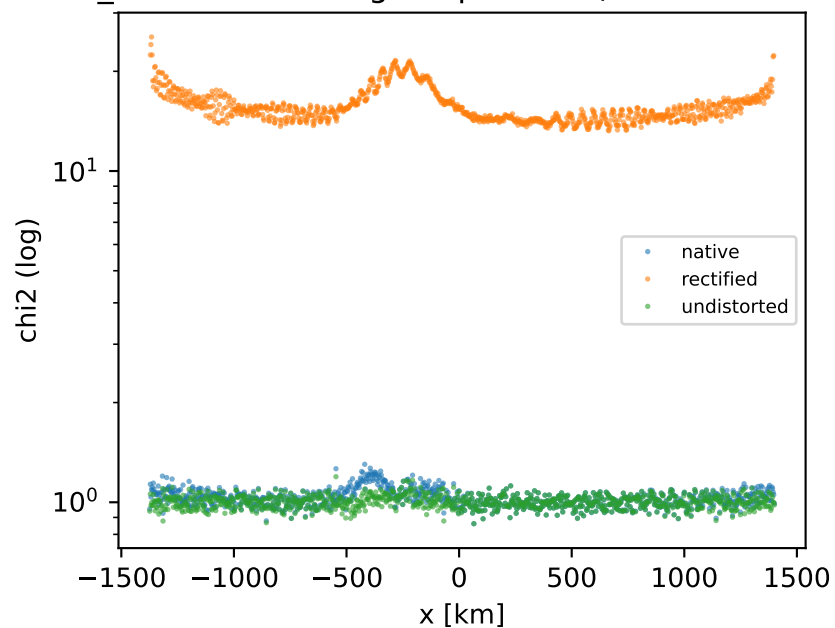
h2o bias vs along-slit position



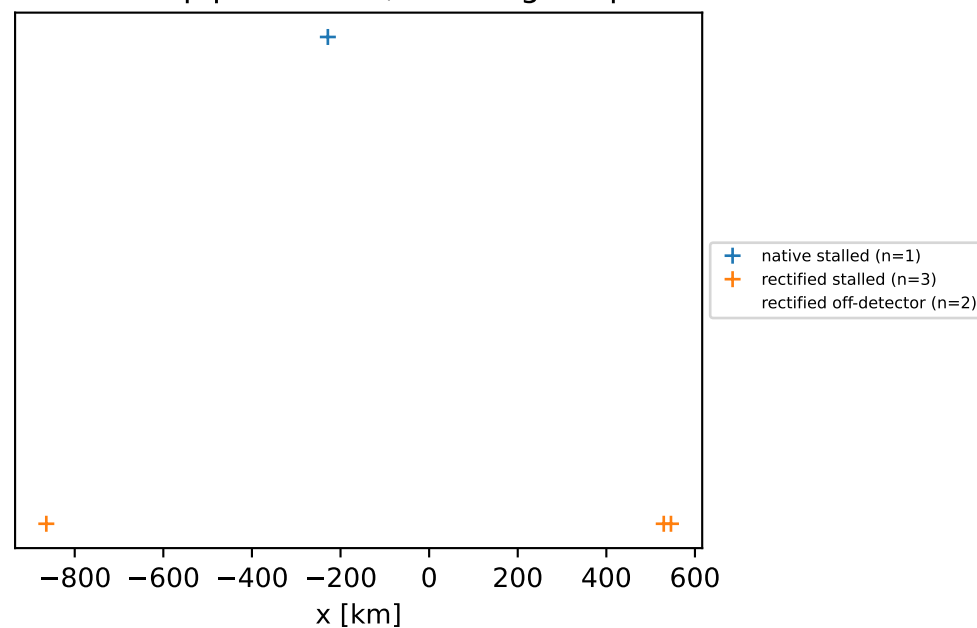
p_surface bias vs along-slit position



chi2_reduced vs along-slit position (outliers excluded)

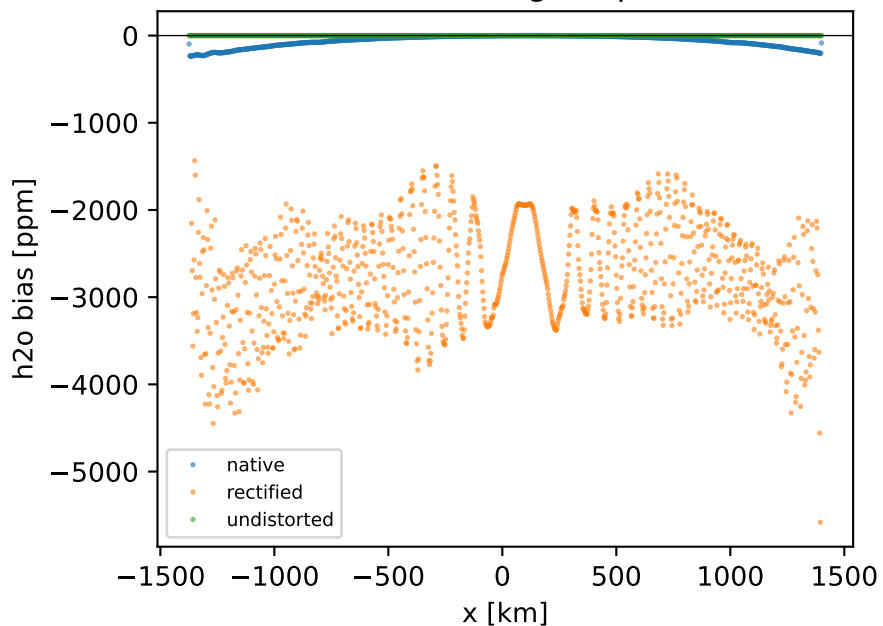


Where each pipeline fails, vs along-slit position

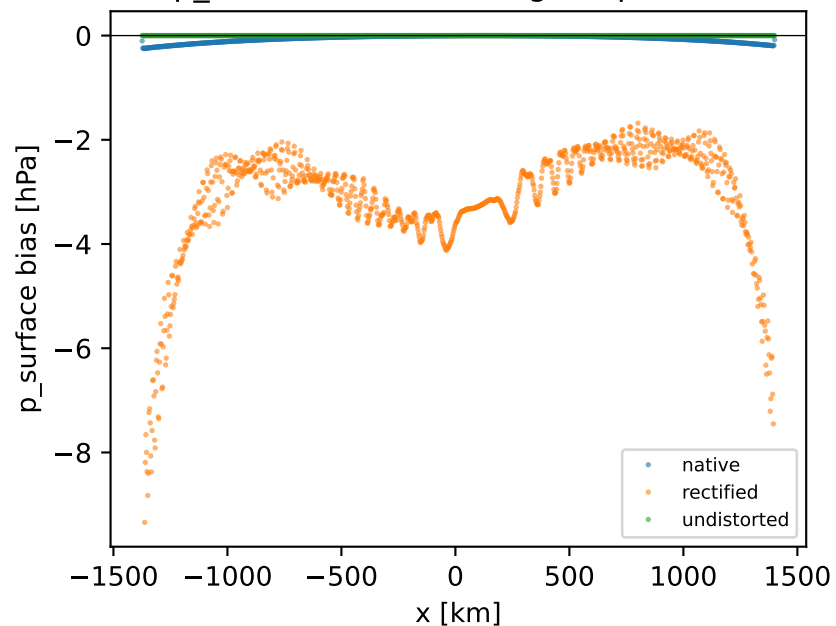


Joint FPA0 -- uniform, no noise (native/rectified/undistorted)

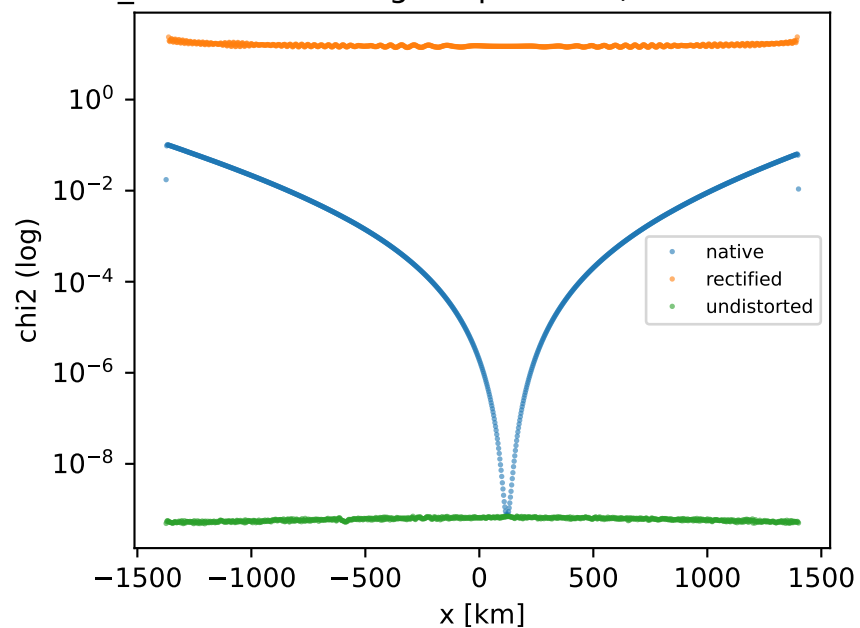
h2o bias vs along-slit position



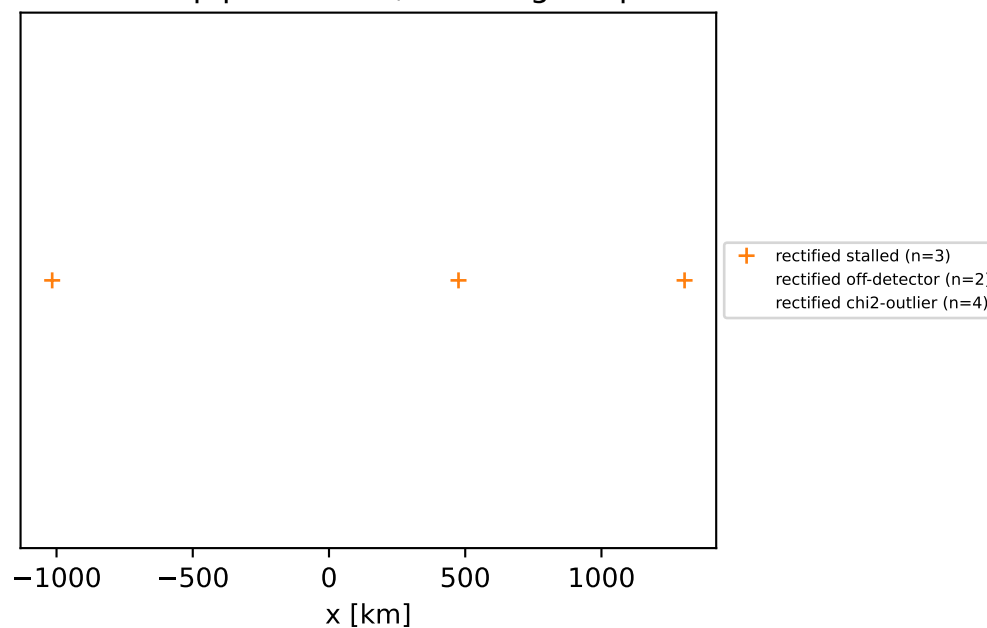
p_surface bias vs along-slit position



chi2_reduced vs along-slit position (outliers excluded)

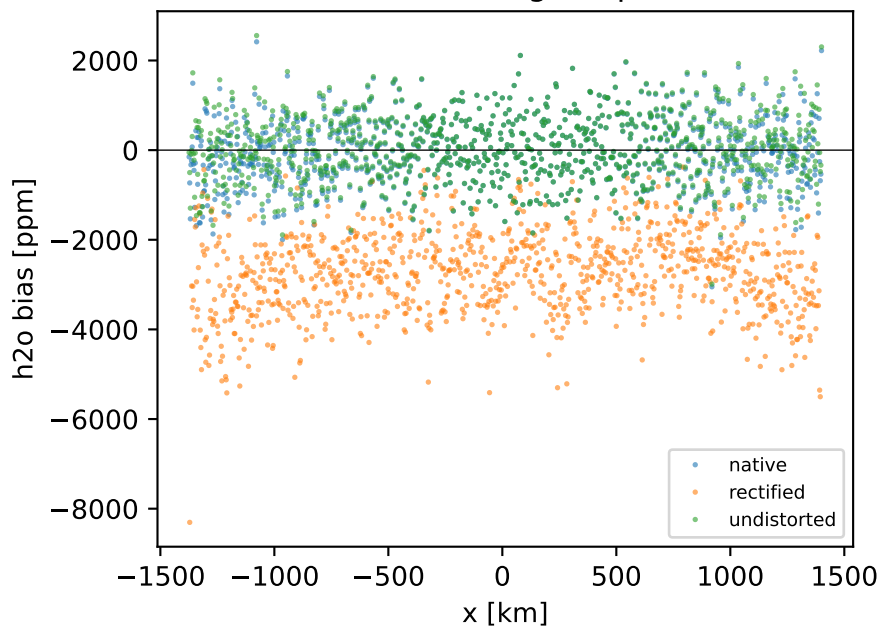


Where each pipeline fails, vs along-slit position

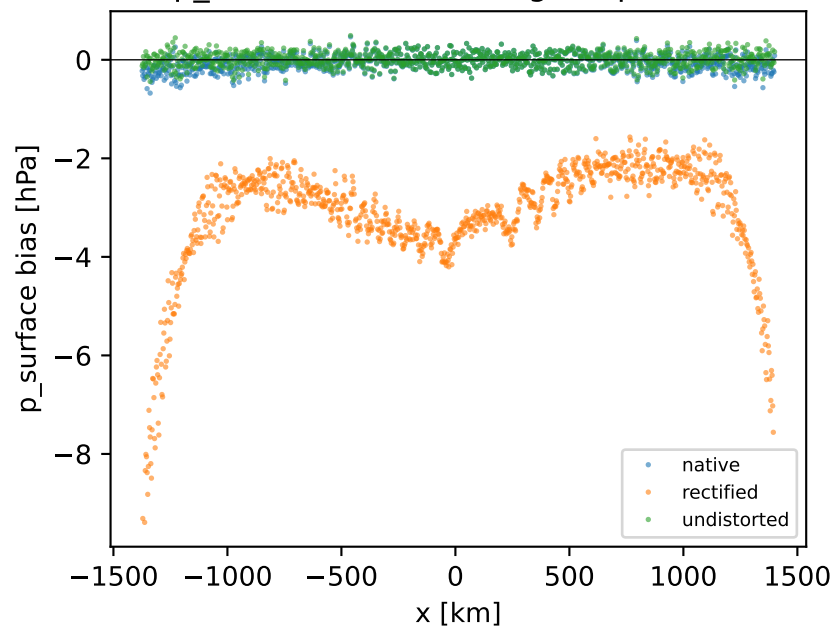


Joint FPA0 -- uniform, noise (native/rectified/undistorted)

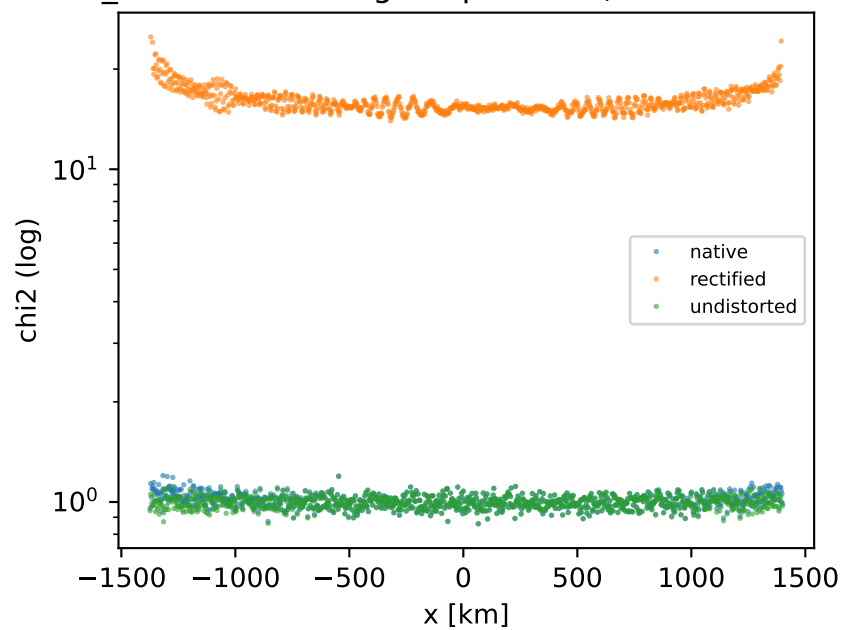
h2o bias vs along-slit position



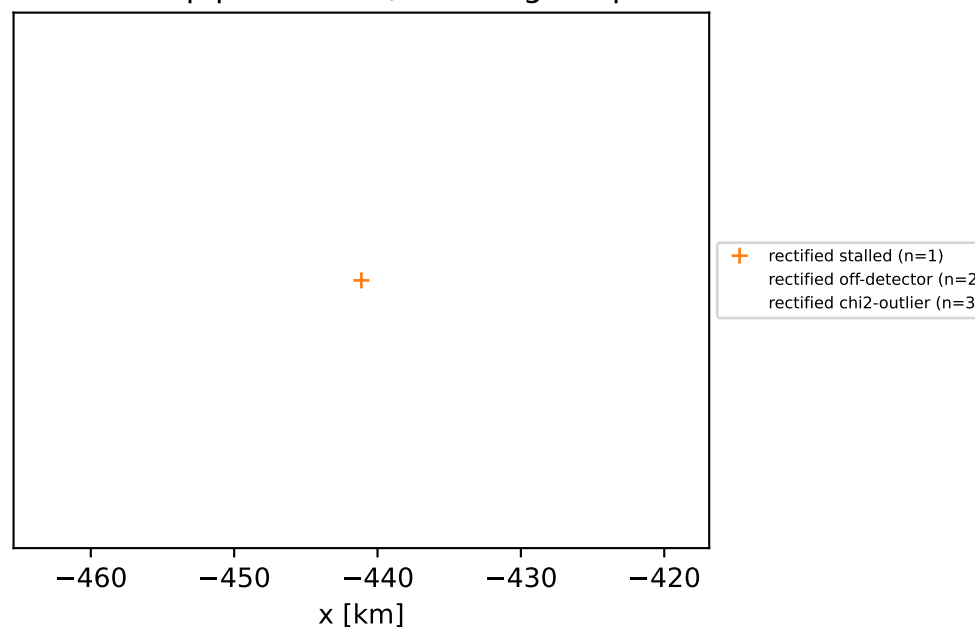
p_surface bias vs along-slit position



chi2_reduced vs along-slit position (outliers excluded)

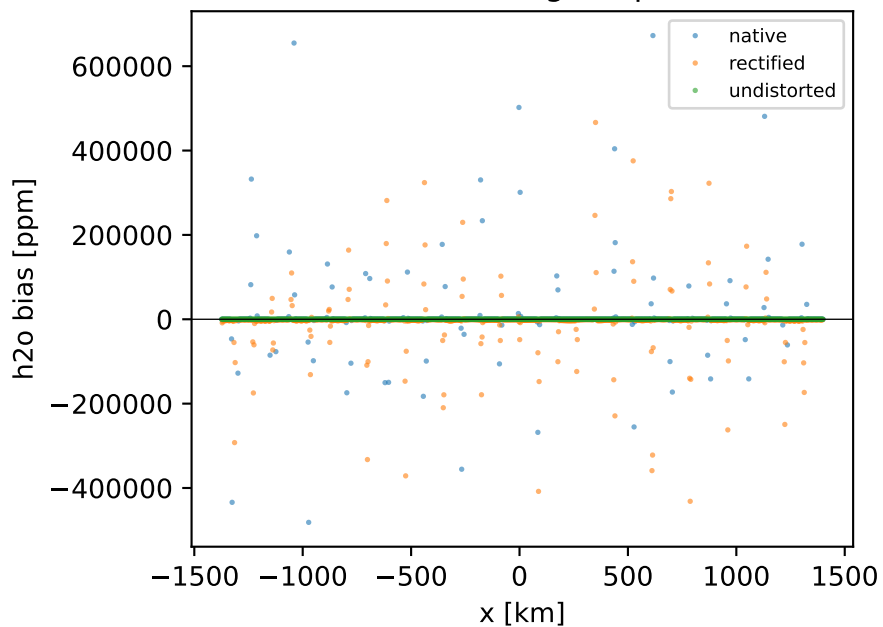


Where each pipeline fails, vs along-slit position

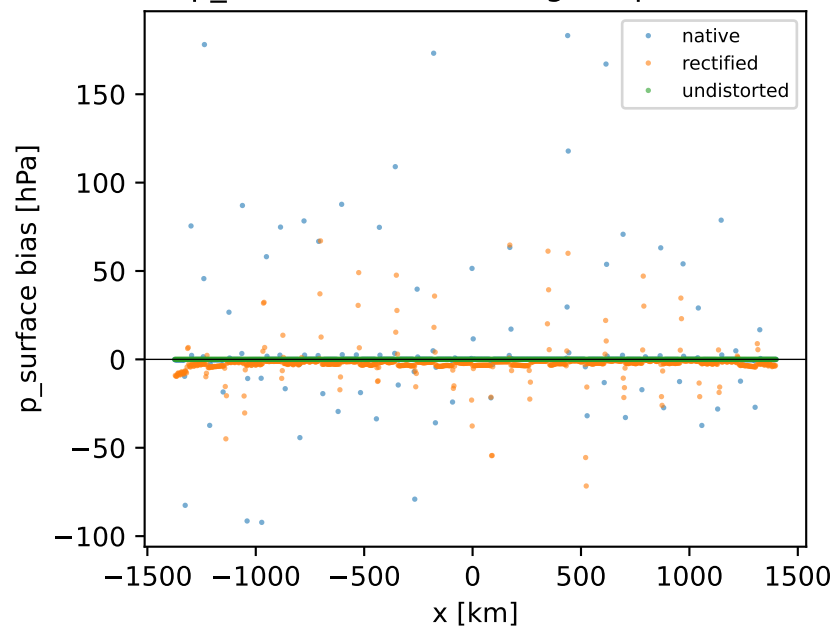


Joint FPA0 -- barcode, no noise (native/rectified/undistorted)

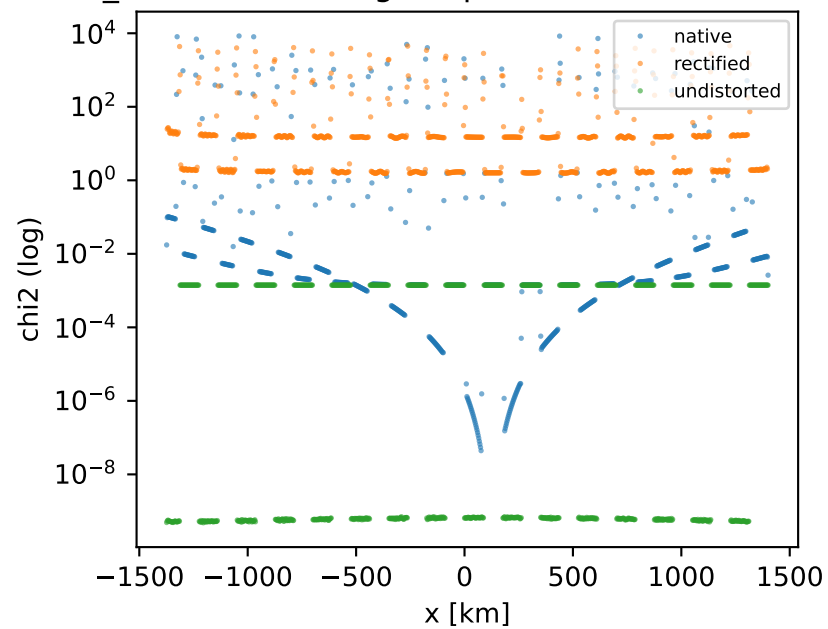
h2o bias vs along-slit position



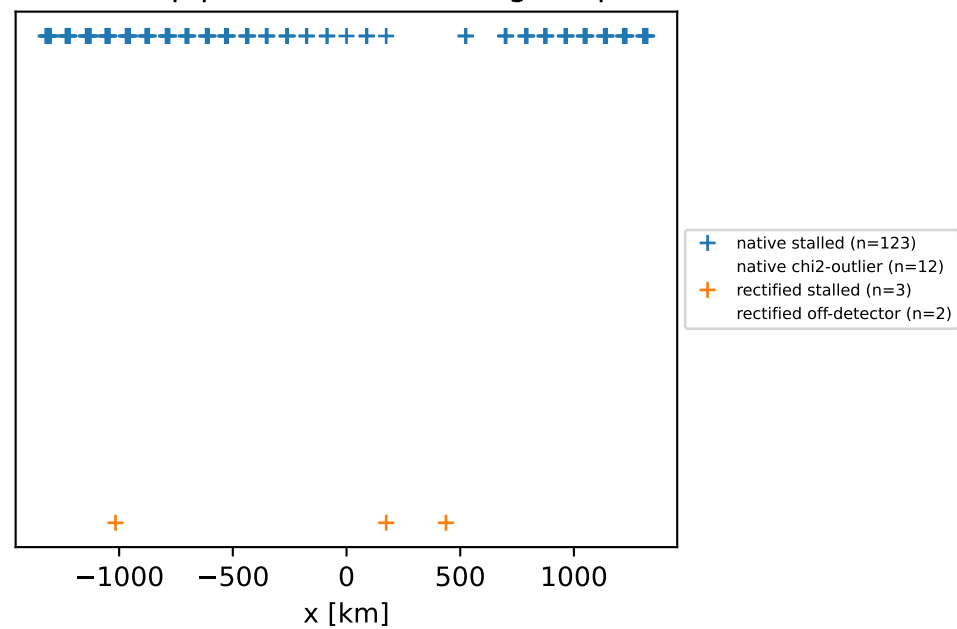
p_surface bias vs along-slit position



chi2_reduced vs along-slit position (outliers excluded)

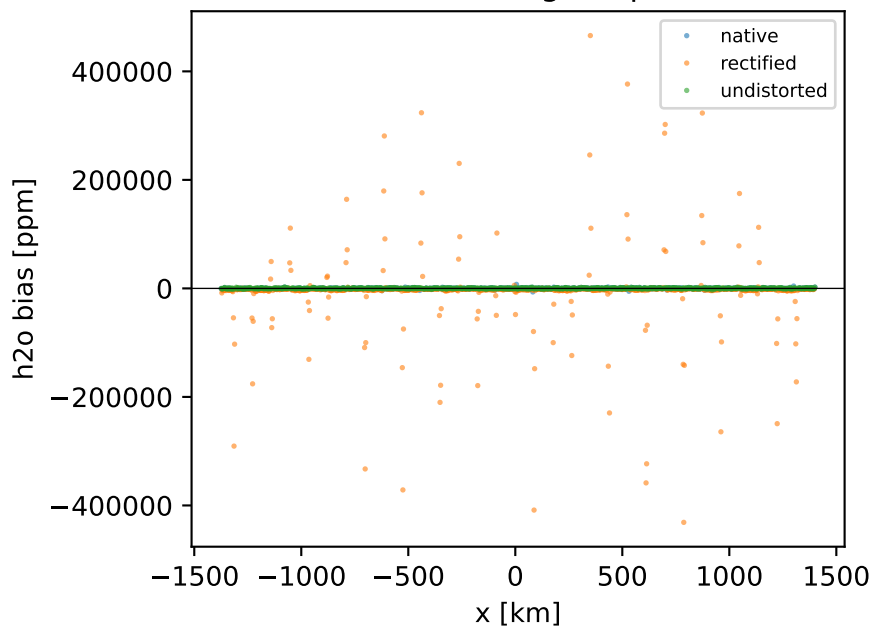


Where each pipeline fails, vs along-slit position

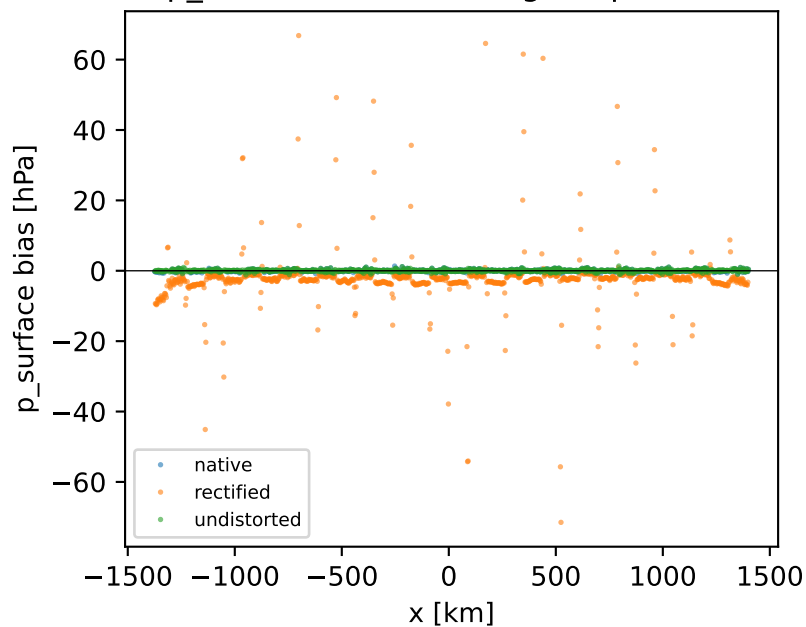


Joint FPA0 -- barcode, noise (native/rectified/undistorted)

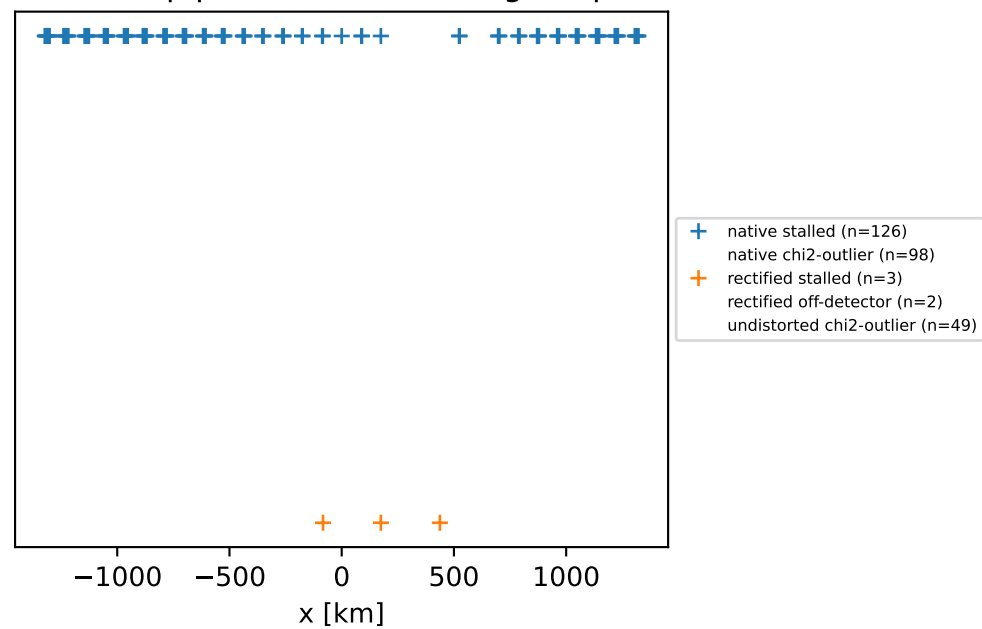
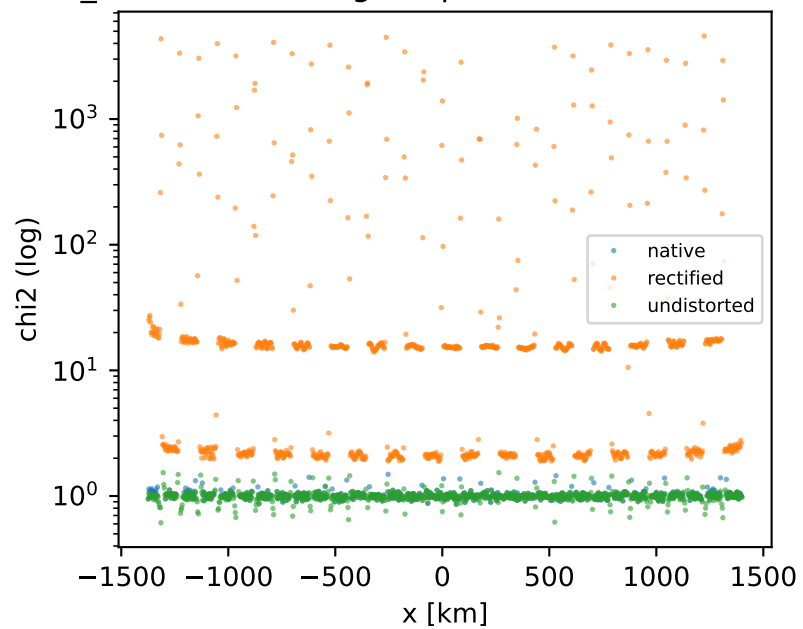
h2o bias vs along-slit position



p_surface bias vs along-slit position



chi2_reduced vs along-slit position (outliers excluded) Where each pipeline fails, vs along-slit position



FPA0 -- robust bias statistics across scenes/noise (native/rectified/undistorted)

scene	noise	quantity	pipeline	n	mean	std	median	MAD-sig	n_out
realistic	0	h2o	native	1024	+325.4	837.1	+321.7	228.1	103
realistic	0	p_surface	native	1024	+0.007536	0.368	-0.02736	0.09554	101
realistic	0	h2o	rectified	1016	-2502	679.8	-2454	677.7	0
realistic	0	p_surface	rectified	1016	-3.311	1.354	-2.922	0.8737	0
realistic	0	h2o	undistorted	1024	+274.9	327.8	+303.6	125.4	0
realistic	0	p_surface	undistorted	1024	+0.01744	0.04572	+0.03661	0.01484	81
realistic	1	h2o	native	1023	+361.8	1111	+343.3	965.2	0
realistic	1	p_surface	native	1023	+0.01312	0.403	-0.01739	0.2276	0
realistic	1	h2o	rectified	1019	-2454	940.2	-2382	939.1	0
realistic	1	p_surface	rectified	1019	-3.31	1.36	-2.929	0.8826	0
realistic	1	h2o	undistorted	1024	+309	791.4	+315.5	786	0
realistic	1	p_surface	undistorted	1024	+0.02263	0.1582	+0.02268	0.1663	0
uniform	0	h2o	native	1024	-63.94	64.88	-40.12	54.85	0
uniform	0	p_surface	native	1024	-0.06306	0.06404	-0.0396	0.05334	0
uniform	0	h2o	rectified	1015	-2764	590.2	-2762	651.9	0
uniform	0	p_surface	rectified	1015	-3.137	1.113	-2.871	0.8026	1
uniform	0	h2o	undistorted	1024	-0.02246	0.0019	-0.02239	0.001962	0
uniform	0	p_surface	undistorted	1024	-3.819e-05	6.707e-06	-4.015e-05	5.208e-06	0
uniform	1	h2o	native	1024	-30.34	740.2	-48.84	743	0
uniform	1	p_surface	native	1024	-0.05818	0.1628	-0.05352	0.1656	0
uniform	1	h2o	rectified	1018	-2725	905.4	-2682	890.7	0
uniform	1	p_surface	rectified	1018	-3.144	1.139	-2.917	0.7915	2
uniform	1	h2o	undistorted	1024	+33.58	734.3	+26.68	732.7	0
uniform	1	p_surface	undistorted	1024	+0.004878	0.1492	+0.00206	0.1572	0
barcode	0	h2o	native	889	+2558	5.965e+04	-16.81	24.35	133
barcode	0	p_surface	native	889	+1.399	17.56	-0.01871	0.0271	126
barcode	0	h2o	rectified	1019	-3480	5.208e+04	-1782	1328	100
barcode	0	p_surface	rectified	1019	-2.043	7.99	-2.241	1.538	49
barcode	0	h2o	undistorted	1024	+0.0003932	0.02257	+0.02136	0.004384	504
barcode	0	p_surface	undistorted	1024	-1.942e-05	1.997e-05	-1.418e-05	2.927e-05	0
barcode	1	h2o	native	800	-11.91	904.9	-28.56	740.5	2
barcode	1	p_surface	native	800	-0.01978	0.3325	-0.03492	0.2508	0
barcode	1	h2o	rectified	1019	-3494	5.208e+04	-1725	1487	97
barcode	1	p_surface	rectified	1019	-2.036	8.004	-2.241	1.506	49
barcode	1	h2o	undistorted	975	+36.75	723.4	+36.3	710.6	0
barcode	1	p_surface	undistorted	975	+0.0223	0.3125	+0.001413	0.2327	0